

SINDHUJA R

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sindhujarajagopalan.vercel.app

PROFILE

Electronics and Communication Engineering graduate transitioning fully into **software development, Artificial Intelligence, and Data Science** with a proven ability to build end-to-end products: from a quantum-ML plant disease classifier in Python and IBM Qiskit, to a deployed personal portfolio in React.js, to an Android control app built with MIT App Inventor. Proficient in Python, Java, and MySQL, with hands-on exposure to UI/UX design through Figma. Comfortable working across the full stack from writing logic to designing interfaces and driven by a desire to solve real-world problems through clean, purposeful software.

WORK EXPERIENCE

Project Intern - Qmax Systems India Pvt. Ltd., Chennai

Dec 2024 – Jan 2025

- Diagnosed circuit faults in embedded systems using oscilloscopes and logic analysers, reducing debug cycle time by identifying root causes at the hardware level.
- Implemented and validated real-time electronics applications on production-grade hardware under direct industry mentorship, bridging the gap between academic lab setups and live systems.

EDUCATION

B.Tech. Electronics and Communication Engineering

CGPA: 7.3587

SASTRA Deemed University, Thanjavur

Higher Secondary Education

Grade 10: 85.6% Grade 12: 89%

Sri Shanmuka Matric Higher Secondary School, Mannargudi

(2019–2022)

SKILLS

Programming Languages: Python, Java, MySQL

AI / Data Science: NumPy, Pandas, Qiskit, IBM Quantum services

Tools & Technologies: Figma (Mobile UI/UX), MATLAB, MIT App Inventor, Arduino IDE

Soft Skills: Cross-functional communication, Design leadership, Structured problem-solving, Deadline management

PROJECTS

AgriQuantum AI - Quantum Computing-Based Plant Disease Prediction System | *Python, Qiskit, IBM Quantum*

- Built a web application using IBM Qiskit and IBM Quantum cloud simulators to process scanned leaf images and classify plant diseases with quantum-enhanced algorithms.
- Designed and tuned quantum circuits for multi-class disease classification, outputting specimen name, disease type, and severity level in real time through an intuitive web interface.
- Bridged quantum computing and practical agriculture the system reduced manual diagnostic effort by providing instant, image-based disease insights directly to growers. ([GIT](#))

Bluetooth - Controlled Industrial Manipulator Robotic Arm | *Arduino, HC-05, MIT App Inventor*

- Engineered a 4-DOF robotic arm for wireless pick-and-place operations, integrating Arduino UNO/NANO microcontrollers, HC-05 Bluetooth module, and servo motors.
- Developed a custom Android control application using MIT App Inventor, achieving reliable full-range control within a 10 m radius via UART communication protocol. ([GIT](#))

Personal Portfolio Website | *HTML, CSS, JavaScript, Vercel*

- Developed a modern, responsive portfolio website to showcase technical projects, skills, certifications, and professional experience using HTML, CSS, and JavaScript.
 - Built interactive user interfaces featuring animated scroll effects, dynamic particle backgrounds, responsive layouts, and project showcase sections to enhance user engagement.
- Deployed and maintained the website on Vercel, applying UI/UX design best practices to ensure accessibility, visual consistency, and cross-device compatibility. ([GIT](#))

COURSES & WORKSHOPS

- Python for Data Science - NPTEL, IIT Madras
- UI/UX Design with Figma - Udemy

LEADERSHIP & EXTRA-CURRICULAR ACTIVITIES

- **Head, Graphics & Designing Team - THETA'26 Fest:** Led a team of designers to conceptualise and execute all visual assets for the department technical festival.
- **Head, Creative & Design Committee - ELCOM DAIS (ECE Association):** Directed creative direction for association events, branding, and student-facing publications.

LANGUAGES

- English - Full Professional Proficiency
- Tamil - Native Speaker